

Consumer Behavior Analytics Dashboard

Power BI dashboard analyzing **3,900 customer transactions** — uncovering purchasing patterns, revenue drivers, and customer segments.

POWER BI

PYTHON PREPROCESSING

RETAIL ANALYTICS



Made with GAMMA

Project Objective

The Challenge

Retailers generate massive transactional data daily — but need tools to extract actionable insights.

The Solution

Interactive BI dashboard summarizing customer behavior, sales performance, and segmentation.

The Process

Python data cleaning → Power BI visualization → management-ready insights.



Dataset at a Glance

3,900

Customers

18

Variables

What's Inside

- **Demographics:** Age, Gender, Location
- **Products:** Category, Item, Color, Size
- **Behavior:** Frequency, Shipping Type
- **Engagement:** Subscription, Discounts, Promo Codes
- **Financials:** Purchase Amount, Review Rating

Dashboard Overview

3 KPI Cards

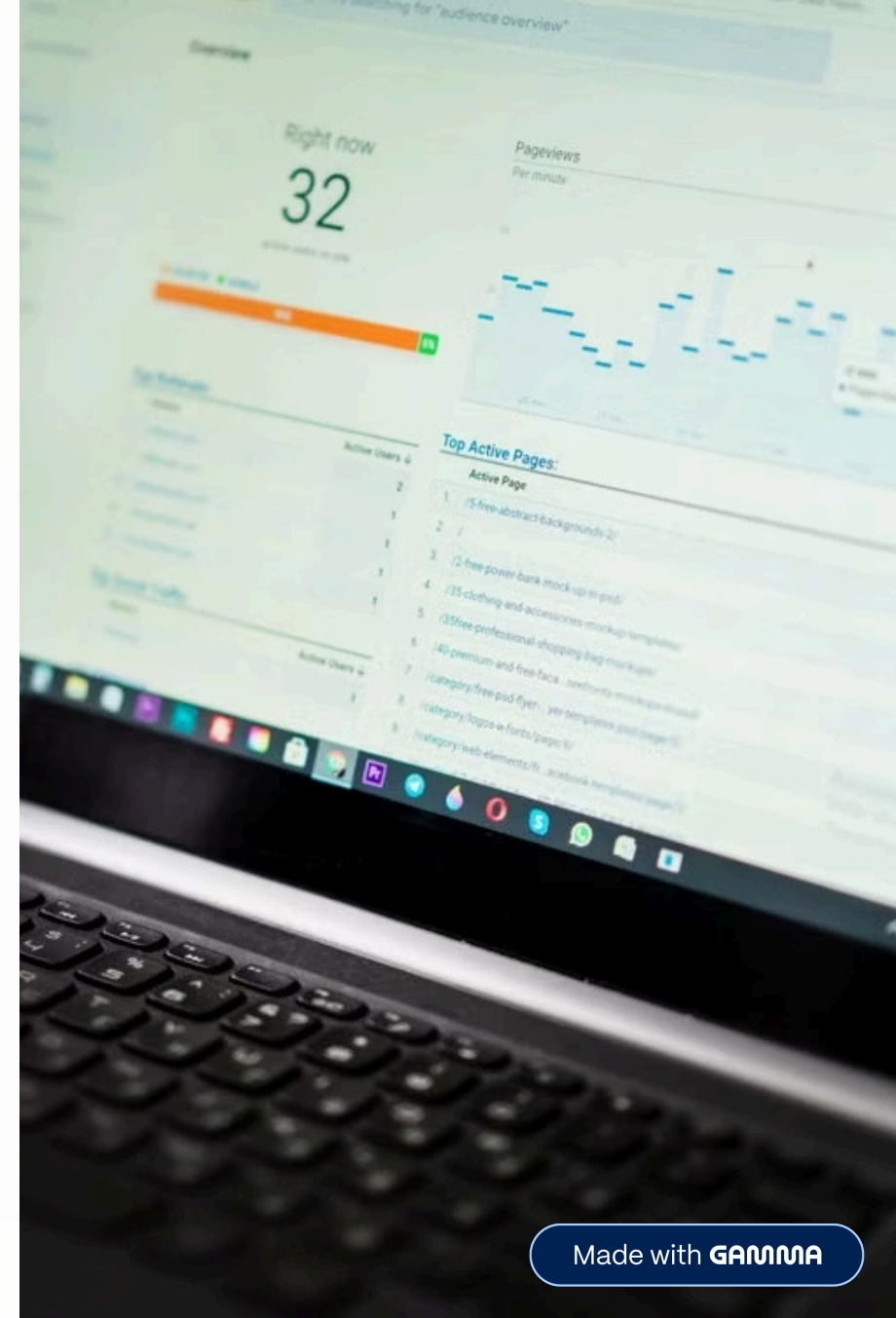
Customers, Avg. Purchase, Avg. Rating

4 Charts

Revenue & Sales by Category & Age Group

4 Slicers

Subscription, Gender, Category, Shipping



Key Performance Indicators

3.9K

Customers

Total transactions in dataset

\$59.76

Avg. Purchase

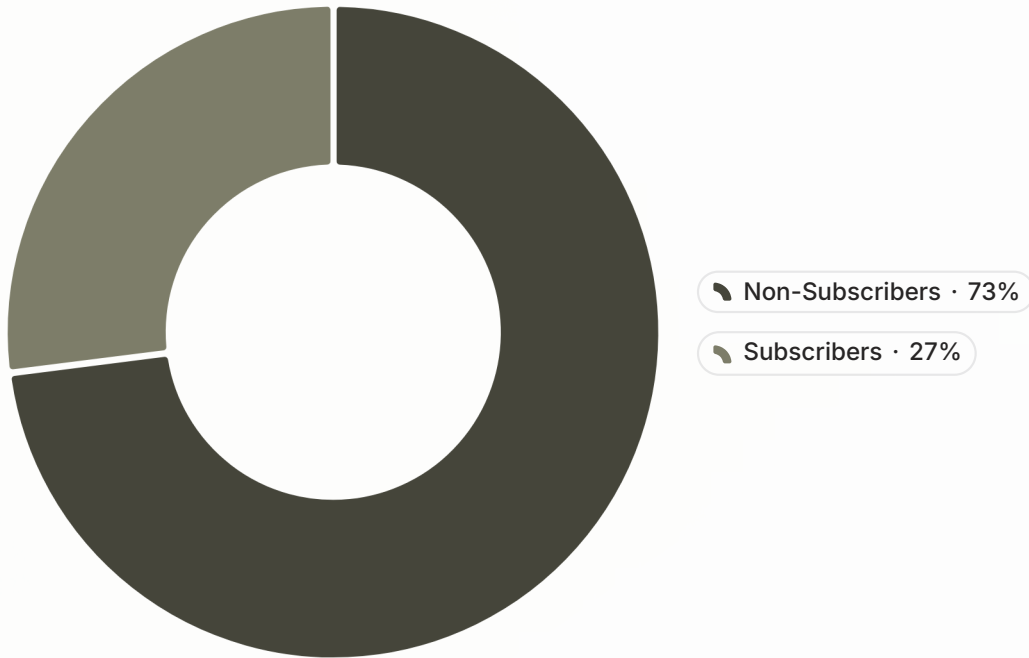
Per transaction benchmark

3.75

Avg. Rating

Out of 5 – generally satisfied

Customer Subscription Analysis

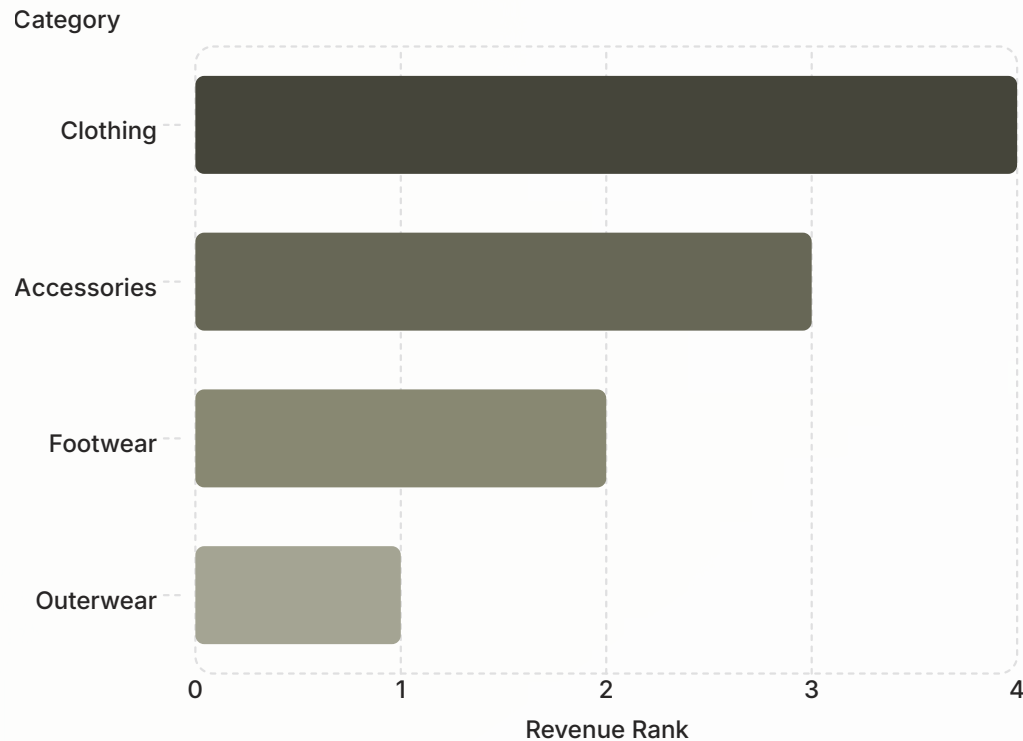


The Opportunity

Only **27%** of customers are subscribed — a significant gap.

- ❏ Expanding loyalty programs could improve retention and encourage repeat purchases.

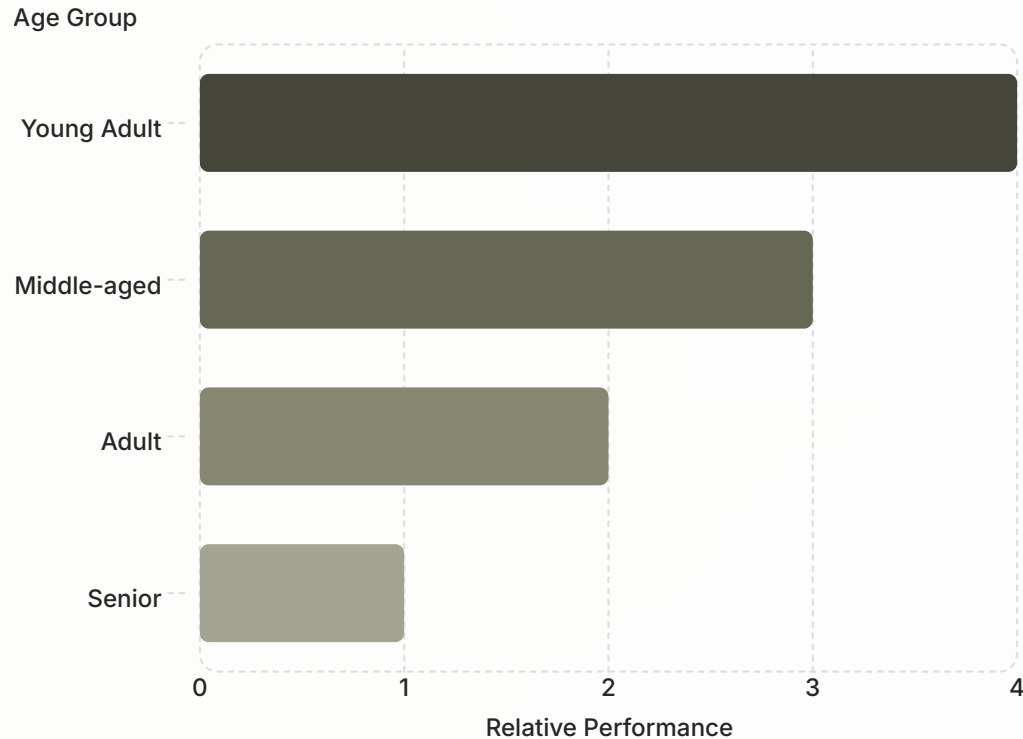
Revenue & Sales by Category



Key Findings

- **Clothing** – highest revenue & sales volume
- **Accessories** – strong second place
- **Outerwear** – lowest performance, needs attention

Revenue & Sales by Age Group



Most Valuable Segment

Young Adults lead in both revenue and purchase frequency.

- Younger consumers are higher-spending *and* more frequent shoppers — ideal for promotions and new launches.

Interactive Dashboard Features



Subscription Status

Filter by subscriber vs. non-subscriber behavior



Gender

Compare purchasing patterns across genders



Product Category

Isolate performance by product type



Shipping Type

Analyze preferences from Standard to Next Day Air

Business Recommendations



Grow Subscriptions

Only 27% subscribed — boost loyalty benefits



Prioritize Clothing

Top revenue driver — expand inventory & promotions



Target Young Adults

Highest spenders — focus digital campaigns here



Fix Outerwear

Lowest performer — review pricing & assortment



Boost Satisfaction

3.75/5 rating — improve quality & delivery